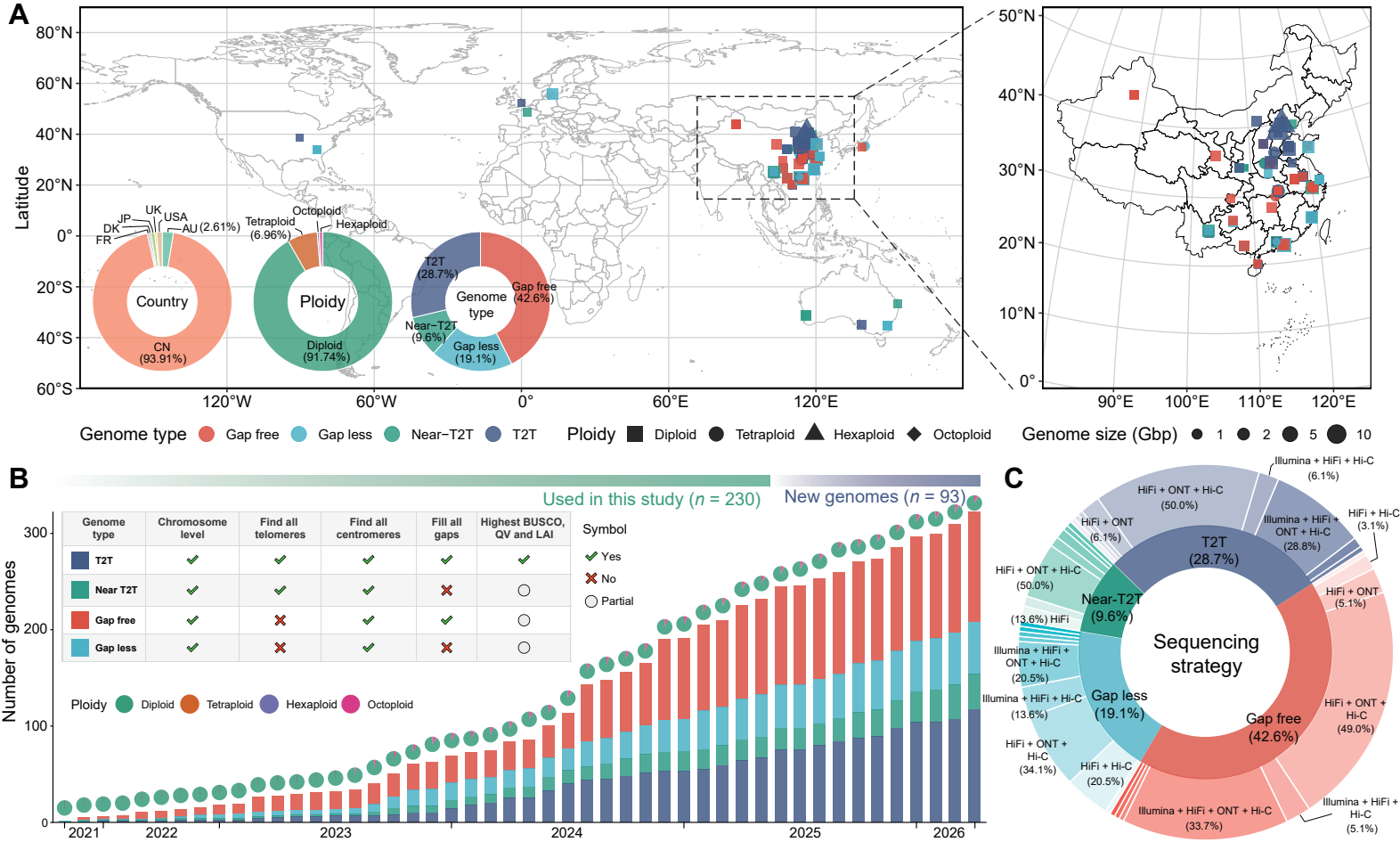

A
Latitude
80°N
60°N
40°N
20°N
0°
20°S
40°S
60°S
120°W
60°W
0°
60°E
120°E
50°N
40°N
30°N
20°N
10°N
0°
90°E
100°E
110°E
120°E
Country
CN (93.91%)
JP
UK
USA
AU (2.61%)
DK
FR
Ploidy
Diploid (91.74%)
Tetraploid (6.96%)
Octoploid
Hexaploid
Genome type
Gap free (42.6%)
Gap less (19.1%)
Near−T2T (9.6%)
T2T (28.7%)
Genome type
Gap free
Gap less
Near−T2T
T2T
Ploidy
Diploid
Tetraploid
Hexaploid
Octoploid
Genome size (Gbp)
1
2
5
10
B
Used in this study (n = 230)
New genomes (n = 93)
Genome type
Chromosome level
Find all telomeres
Find all centromeres
Fill all gaps
Highest BUSCO, QV and LAI
T2T
Near T2T
Gap free
Gap less
Symbol
Yes
No
Partial
Ploidy
Diploid
Tetraploid
Hexaploid
Octoploid
Number of genomes
300
200
100
0
2021
2022
2023
2024
2025
2026
C
Sequencing strategy
T2T (28.7%)
Near-T2T (9.6%)
Gap less (19.1%)
Gap free (42.6%)
Illumina + HiFi + Hi-C (6.1%)
HiFi + ONT + Hi-C (50.0%)
Illumina + HiFi + ONT + Hi-C (28.8%)
HiFi + Hi-C (3.1%)
HiFi + ONT (6.1%)
HiFi + ONT + Hi-C (50.0%)
HiFi + ONT (5.1%)
(13.6%) HiFi
Illumina + HiFi + ONT + Hi-C (20.5%)
Illumina + HiFi + Hi-C (13.6%)
HiFi + ONT + Hi-C (49.0%)
HiFi + ONT + Hi-C (34.1%)
HiFi + Hi-C (20.5%)
Illumina + HiFi + ONT + Hi-C (33.7%)
Illumina + HiFi + Hi-C (5.1%)